

AMCS 355 Homework

due in class Feb. 15th

Consider the advection equation

$$\begin{aligned}u_t &= -u_x \\ u(x, 0) &= u_0(x) \\ u(0, t) &= u(1, t)\end{aligned}$$

posed on the interval $x \in [0, 1]$, with periodic boundary conditions. The exact solution of this problem is dissipative in the 1-, 2-, and max-norms. Use the logarithmic norm to determine whether dissipativity holds for the solution with each of the following semi-discretizations:

1. Upwind:

$$u'_i(t) = -\frac{u_i - u_{i-1}}{\Delta x}.$$

2. Centered:

$$u'_i(t) = -\frac{u_{i+1} - u_{i-1}}{2\Delta x}.$$

Note: for the 2-norm, you will need to compute eigenvalues. This can be done using von Neumann analysis, or you can just compute them numerically for some example values of Δx if you prefer.